Q1. Smart Hospital Patient Registration System

CODE

```
import java.util.*;
public class Main{
    static class Patient{
        String id,name,mobile,dept,doctor;
        Patient(String id,String name,String mobile,String dept,String doctor){
            this.id=id;
            this.name=name;
            this.mobile=mobile;
            this.dept=dept;
            this.doctor=doctor;
        }
    }
    private static String toProperCase(String str){
        String[] words = str.trim().split("\\s+");
        StringBuilder sb = new StringBuilder();
        for(String w:words){
            if(!w.isEmpty())sb.append(Character.toUpperCase(w.charAt(0))).append(w.substring(1).toLowerCase()).append(" ");
        }
        return sb.toString().trim();
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        List<Patient>list=new ArrayList<>();
        Map<String,Patient> map=new HashMap();
        while(sc.hasNextInt()){
            int choice=sc.nextInt();
            sc.nextLine();
            if(choice==1){
                String id = sc.nextLine().trim();
                String name = sc.nextLine().trim();
                String mobile = sc.nextLine().trim();
                String dept = sc.nextLine().trim();
                String doctor = sc.nextLine().trim();
                if(map.containsKey(id)){System.out.println("Patient ID already exists.");continue;}
                if(!name.matches("[a-zA-Z\\s]+")){System.out.println("Invalid patient Name.");continue;}
                if(!mobile.matches("\\d{10}")){System.out.println("Invalid Mobile Number.");continue;}
                Patient p = new Patient(id,toProperCase(name),mobile,dept.toUpperCase(),doctor);
                list.add(p);
                map.put(id,p);
                System.out.println("Patient Registered Successfully.");
            }else if(choice==2){
                String searchId = sc.nextLine().trim();
                if(map.containsKey(searchId)){
                    Patient p = map.get(searchId);
                        System.out.println("Patient Detail\nID : "+p.id+"\nName : "+p.name+"\nMobile : "+p.mobile+"\nDepartment : "+p.
dept+"\nDoctor : "+p.doctor);
            }else if(choice==3){
                System.out.println("---------------------------------------\nID    Name   Department\n-------------------------------
----");
                 for(Patient p : list)System.out.printf("%-8s%-14s%s\n",p.id,p.name,p.dept);
            }else if(choice==4){
                System.out.println("Department Count");
                Map<String,Integer>counts=new LinkedHashMap<>();
                for(Patient p : list)counts.put(p.dept,counts.getOrDefault(p.dept,0)+1);
                for(var e : counts.entrySet()) System.out.println(e.getKey()+":"+e.getValue());
            }else if(choice==5)
                    break;
            }
        }
    }
}
```

INPUT

```
P101
Rahul Kumar
9887578562
Cardiology
Dr.Arun
1
P102
Priya
9876543211
Ortho
Dr.meena
3
4
2
P101
5
```

OUTPUT

```
(no output)
```

Q2. Online Library Book Management System

CODE

```
import java.util.*;
public class Main{
    static class Book{
        String id,title,author,category,publisher;
        Book(String id,String title,String author,String category,String publisher){
            this.id=id;
            this.title=title;
            this.author=author;
            this.category=category;
            this.publisher=publisher;
        }
    }
    private static String toTitleCase(String str){
        String[] words = str.trim().split("\\s+");
        StringBuilder sb = new StringBuilder();
        for(String w:words){
            if(!w.isEmpty()){
                sb.append(Character.toUpperCase(w.charAt(0)))
                  .append(w.substring(1).toLowerCase())
                  .append(" ");
            }
        }
        return sb.toString().trim();
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        List<Book> list = new ArrayList<>();
        Map<String>,Book>map=new HashMap<>();
        Set<String> authors=new LinkedHashSet<>();
        while(sc.hasNextInt()){
            int choice=sc.nextInt();
            sc.nextLine();
            if(choice==1){
                String id = sc.nextLine().trim();
                String title = sc.nextLine().trim();
                String author = sc.nextLine().trim();
                String category = sc.nextLine().trim();
                String publisher = sc.nextLine().trim();
                if(map.containsKey(id)){System.out.println("Book ID already exists.");continue;}
                if(publisher.isEmpty()){System.out.println("Publisher cannot be empty.");continue;}
                Book b = new Book(id,toTitleCase(title),author,category.toUpperCase(),publisher);
                list.add(b);map.put(id,b);authors.add(author);
                System.out.println("Book Added Successfully.");
            }else if(choice==2){
                String searchId=sc.nextLine().trim();
                if(map.containsKey(searchId)){
                    Book b = map.get(searchId);
                    System.out.println("Book Details\nBook ID : "+b.id+"\nTitle : "+b.title+"\nAuthor : "+b.author+"\nCategory : "+b.c
ategory+"\nPublisher : "+b.publsher);
                }else System.out.println("Book not Available";)
            }else if(choice==3){
                System.out.println("--------------------\nBook ID Title  Author\n-----------------------");
                for(Book b : list)System.out.printf("%-10s%-23s%s\n",b.id,b.title,b.author);
            }else if(choice==4){
                System.out.println("Unique Authors");
                for(String a:authors)System.out.println(a);
            }else if(choice==5){
                System.out.println("Category-wise count");
                Map<String,Integer> counts = new LinkedHashMap<>();
                for(Book b : list)counts.put(b.category,counts.getOrDefault(b.category,0)+1);
                for(var e : counts.entrySet())System.out.println(e.getKey()+":"+e.getValue());
            }else if(Choice==6)break;
```

```
        }
    }
}
```

OUTPUT

```
(no output)
```